

PROJECT REPORT
ON
BMI CALCULATOR

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE
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In partial fulfilment of the requirements for the award of the

Degree of Bachelor of Technology

in

INFORMATION SCIENCE AND ENGINEERING

by

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**NMAM INSTITUTE
OF TECHNOLOGY**

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**NMAM INSTITUTE
OF TECHNOLOGY**

Department of Information Science & Engineering

CERTIFICATE

This is to certify that Ms. Adithi Pai S(NNM22IS08) and Ms. Hemani Rao (NNM22IS057) of II-year B.Tech., a bonafide students of NMAM Institute of Technology, Nitte, has carried out project on “**BMI CALCULATOR**” as part of the **Introduction to Data Science (IS1102-1)** course during 2023-24, fulfilling the partial requirements for the award of degree of Bachelor of Technology in Information Science and Engineering at NMAM Institute of Technology, Nitte.

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PROJECT EVALUATION MARKS

Evaluation Component	Implementation (20 Marks)	Presentation (10 Marks)	Question and Answer (10 Marks)	Organization of the report (10 Marks)	Total (50 Marks)
Marks					

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Signature of Course Instructor

PROJECT EVALUATION RUBRICS

Evaluation Component	Evaluation Levels			
	Excellent (16-20)	Good (10 – 15)	Satisfactory (1-9)	Poor 0
Implementation	The project exceeds expectations with outstanding performance, creativity, and attention to detail.	The project meets expectations and demonstrates a solid understanding of the criteria.	The project partially meets expectations but requires improvement in certain areas.	The project does not meet expectations and requires significant improvement
Presentation (10 Marks)	Excellent (9-10) Uses graphics that explain and reinforce text and presentation. Refers to slides to make points, engaged with audience Length and Pace of presentation is Appropriate and well-paced throughout	Good (5 – 8) Uses graphics that explain text and presentation. Refers to slides to make points, eye contact majority of time Length and Pace of presentation is Adequate and most of the seminar well-paced	Satisfactory (1-4) Uses graphics that relate to text and presentation. Refers to slides to make points; occasional eye contact Length and Pace of presentation is Short OR long. Rushed OR dragging in parts	Poor 0 Uses graphics that rarely support text and Presentation. Reads most slides. No or just occasional eye Very Short. Length and Pace of presentation is Rushed OR dragging
Question and Answer (10 Marks)	Demonstrated full knowledge. Sufficient for understanding and exceptionally answered for all the questions	Demonstrated knowledge Sufficient for understanding and answered for all the questions	Demonstrated some knowledge. insufficient for understanding and answered for few questions	Does not have grasp of Information and not answered for even one question
Organization of the report (10 Marks)	Well documented content and followed all specified formats	Documented content and followed almost all specified formats	Fair document. Specified formats are not followed	Poor document content. Specified formats are not followed, submitted after deadline

ABSTRACT

Body Mass Index (BMI) stands as a cornerstone in the assessment of an individual's weight status relative to their height. This widely-utilized metric has played a crucial role in public health, clinical settings, and research. Originating as a simple calculation, BMI has evolved into a valuable tool for identifying potential health risks associated with underweight, normal weight, overweight, and obesity. However, its limitations, including a lack of differentiation between muscle and fat mass, have spurred ongoing discussions about its applicability in diverse populations.

This abstract provides an overview of the historical development of BMI, its calculation methodology, and its significance in health assessments. It explores the categorization of BMI and its implications for individual health, disease prevention, and public health initiatives. While BMI remains a valuable screening tool, its shortcomings highlight the need for a more comprehensive approach to body composition analysis. The future scope involves further research into refining BMI assessments, integrating complementary measurements, and considering population-specific norms. This abstract sets the stage for a nuanced understanding of BMI, acknowledging its contributions while emphasizing the evolving landscape of health assessment methodologies.

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CHAPTER 1

INTRODUCTION

Body Mass Index (BMI) is a crucial tool for assessing body composition by examining the relationship between weight and height. This widely-utilized metric serves as a valuable guide for individuals, healthcare professionals, and researchers, offering insights into whether one falls within a healthy weight range or faces potential health risks associated with weight. In this comprehensive guide, we will navigate the origins and development of BMI, exploring its significance in the realm of health and wellness. Whether you seek self-understanding, aim to promote a healthier lifestyle, or are intrigued by the science behind body weight assessments, this resource will delve into BMI's history, calculation methods, and various weight categories, addressing its limitations and introducing alternative measures.

Our journey will not only uncover the evolution of BMI as a screening tool but also highlight its role in disease prevention and emphasize the importance of maintaining a balanced body weight. Whether you are a healthcare professional guiding patients toward healthier living, an individual on a personal wellness journey, or simply curious about the intricacies of body weight assessment, this guide is designed to be your go-to source for all things BMI. Join us as we embark on an enlightening exploration of Body Mass Index, gaining valuable insights into the nuanced relationship between weight, height, and overall health.

CHAPTER 2

IMPLEMENTATION

2.1 Module Explanation

2.1.1 Objective:

BMI is a versatile tool with applications in health assessment, disease risk prediction, population health studies, treatment planning, and resource allocation. It efficiently assesses an individual's weight relative to height, serving as a screening tool for health risks. Correlations with conditions like cardiovascular diseases inform risk prediction, while at the population level, BMI aids in analyzing weight-related health trends for effective public health interventions. Healthcare professionals use BMI for treatment planning, and its data guides resource allocation by identifying populations at higher health risk, enabling targeted interventions and efficient use of resources.

2.1.2 Components:

Base R Functions:

`getwd()`: Retrieves the current working directory.

`print()`: Prints the output.

dplyr Package:

`mutate()`: Used for adding new variables to the dataset (in this case, calculating and adding the BMI variable).

`%>%` (pipe operator): Used to chain together multiple operations in a readable and sequential manner.

`ggplot()` Package- Popular package for creating complex and customizable plots.

Data Reading and Manipulation:

`read.csv()`: Reads a CSV file into a data frame.

`write.csv()`: Writes a data frame to a CSV file.

Visualization:

`hist()`: Creates a histogram for visualizing the distribution of BMI.

`plot()`: Creates a scatter plot for visualizing the relationship between height and weight.

`Pie()`: Creates pie graph to show various categories

`Ggplot()`: Create a bar graph of BMI categories using ggplot

2.1.3 Methodology

The Body Mass Index (BMI) is a simple mathematical formula calculated using an individual's weight and height. The standard formula for BMI is:

$$\text{BMI} = \text{Height (m)}^2 / \text{Weight (kg)}$$

Here's a step-by-step methodology for calculating BMI:

1. Units of Measurement: Ensure that weight is measured in kilograms (kg) and height is measured in meters (m). If measurements are in pounds and inches, convert weight to kilograms by dividing by 2.205 (1 kg = 2.205 lbs) and height to meters by multiplying by 0.0254 (1 inch = 0.0254 meters).

2. Square the Height:

Square the height value obtained in meters (Height (m)^2)

3. Calculate BMI:

Divide the weight in kilograms by the squared height ($\text{Weight (kg)} / \text{Height (m)}^2$).

4. Interpretation:

The resulting BMI value falls into one of several categories:

- ☐ Underweight: BMI less than 2
- ☐ Normal weight: BMI between 2 and 3
- ☐ Overweight: BMI above 5

DATASET:

	A	B	C	D	E	F
1		Height	Weight	Index	BMI	BMI_Category
2	Male	174	96	4	2.229313	Normal
3	Male	189	87	2	1.712356	Underweight
4	Female	185	110	4	2.259683	Normal
5	Female	195	104	3	1.922925	Underweight
6	Male	149	61	3	1.931771	Underweight
7	Male	189	104	3	2.046954	Normal
8	Male	147	92	5	2.993308	Normal
9	Male	154	111	5	3.290636	Normal
10	Male	174	90	3	2.089981	Normal
11	Female	169	103	4	2.535491	Normal
12	Male	195	81	2	1.497663	Underweight
13	Female	159	80	4	2.224816	Normal
14	Female	192	101	3	1.92627	Underweight
15	Male	155	51	2	1.492468	Underweight
16	Male	191	79	2	1.522505	Underweight
17	Female	153	107	5	3.213655	Normal
18	Female	157	110	5	3.137557	Normal
19	Male	140	129	5	4.627346	Normal
20	Male	144	145	5	4.916333	Normal
21	Male	172	139	5	3.303362	Normal
22	Male	157	110	5	3.137557	Normal
23	Female	153	149	5	4.475089	Normal
24	Female	169	97	4	2.387793	Normal
25	Male	185	139	5	2.855418	Normal
26	Female	172	67	2	1.592268	Underweight
27	Female	151	64	3	1.973442	Underweight
28	Male	190	95	3	1.850183	Underweight

CODE:

```
#get and print current working directory
print(getwd())
```

```
# Load the necessary library
library(dplyr)
library(ggplot2)
```

```
#reading a CSV file
data<-read.csv("bmi.csv ")
data
```

```
# Check the structure of the data
```

```
str(data)
```

```
# Calculate BMI
```

```
calculate_bmi <- function(height, weight)
```

```
{
```

```
  # Convert height from inches to meters and weight from pounds to kg.
```

```
  bmi <- (weight*0.45359237) / ((height*0.0254)^2)
```

```
  return(bmi)
```

```
}
```

```
# Apply the calculate_bmi function to the data
```

```
data <- data %>%
```

```
  mutate(BMI = calculate_bmi(Height, Weight))
```

```
# Display the updated data with BMI
```

```
print(data)
```

```
# Create a histogram of BMI
```

```
hist(data$BMI, col = "black", border = "white",
```

```
  main = "Histogram of BMI", xlab = "BMI", ylab = "Frequency")
```

```
# Create a scatter plot
```

```
plot(data$Height, data$Weight, col = data$BMI,
```

```
  main = "Scatter Plot of Height vs. Weight", xlab = "Height (inches)", ylab = "Weight (pounds)")
```

```
# Calculate BMI categories
```

```
data <- data %>%
```

```
  mutate(BMI_Category = cut(BMI, breaks = c(-Inf, 2, 5, Inf), labels = c("Underweight", "Normal",  
"Overweight")))
```

```
# Display the updated data with BMI and categories
```

```
print(data)
```

```
# Save the updated data with BMI and categories to a new CSV file
```

```
write.csv(data, "dataset_with_bmi_and_categories.csv", row.names = FALSE)
```

```
# Create a pie chart of BMI categories
```

```
pie(table(data$BMI_Category), main = "Pie Chart of BMI Categories")
```

```
# Create a bar graph of BMI categories using ggplot2
```

```
ggplot(data, aes(x = BMI_Category)) +  
  geom_bar(fill = "blue", color = "black") +  
  labs(title = "Bar Graph of BMI Categories", x = "BMI Category", y = "Count")
```

CHAPTER 3

RESULTS AND DISCUSSION

This function is defined to calculate the BMI.

It takes two parameters: `weight_kg` (weight in kilograms) and `height_m` (height in meters).

Inside the function, it calculates BMI using the formula: $BMI = \text{weight (kg)} / (\text{height (m)}^2)$.

The result is stored in the `bmi` variable, and then the function returns this value.

Input Values:

The code specifies the weight (154.324 pounds) and height (68.5 inches) as input values.

Calculate BMI:

It calculates the BMI by calling the `calculate_bmi` function with the provided weight and height values and stores the result in the `bmi_result` variable.

Print Result:

It prints the calculated BMI value using `cat()`, rounded to two decimal places, with a message "Your BMI is:".

Interpret the BMI:

The code uses a series of `if` and `else if` statements to interpret the BMI value based on common BMI categories.

If the BMI is less than 2, it prints "You are underweight."

If the BMI is between 2 (inclusive) and 3 (exclusive), it prints "You have a normal weight."

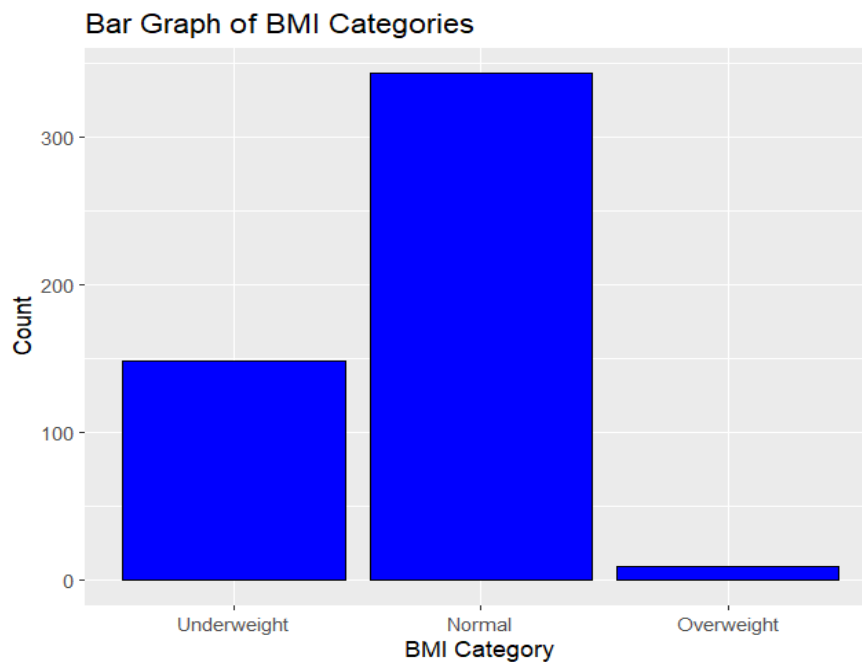
If the BMI is between 3 (inclusive) and 5 (exclusive), it prints "You are overweight."

If none of the above conditions are met, it prints "You are obese."

Overall, this code calculates BMI from weight and height, prints the BMI value, and provides an interpretation of the BMI category (underweight, normal weight, overweight, or obese) based on common ranges.

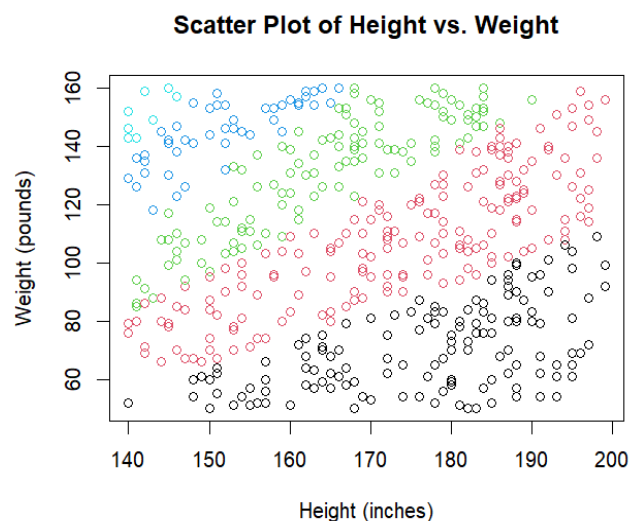
OUTPUT:

Bar Graph of BMI Categories



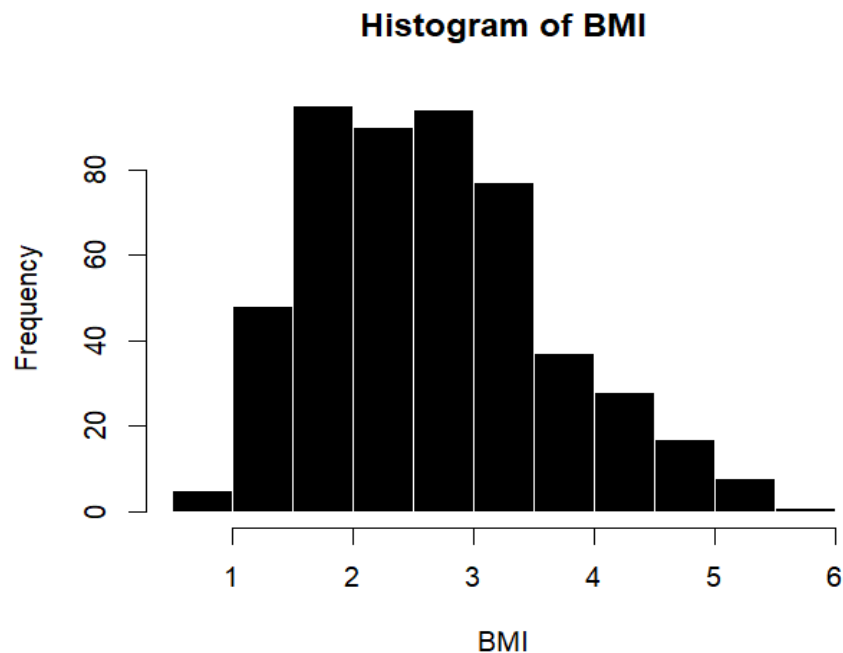
From the above bar graph we can see that in this dataset, the highest number of people fall under the Normal category and the least count in the Overweight category.

Scatter plot of Height vs Weight



The scatter plot depicts the range of values taken for the dataset. The range of height is within 140 to 200 inches and the range for Weight is 40 to 160 pounds.

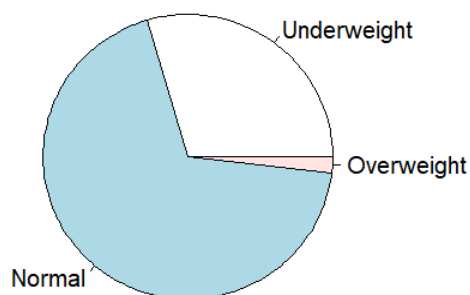
Histogram of BMI



The histogram depicts the number of people that fall under various BMI indexes. Highest frequency is between the BMI Index 1.5-3

Pie chart of BMI Categories

Pie Chart of BMI Categories



The above pie chart depicts the categories(Underweight ,Overweight and Normal).In this dataset maximum people fall under the Normal Category.

CHAPTER 4

CONCLUSION

In conclusion, Body Mass Index (BMI) serves as a valuable tool for assessing body weight in relation to height, providing a quick and accessible metric for health professionals and individuals alike. While widely used, it's crucial to recognize the limitations of BMI, such as its inability to differentiate between muscle and fat mass and its reliance on a simplified height-to-weight ratio.

As we move forward, the future scope of BMI involves continued research to refine and enhance the accuracy of body composition assessments. Integration with other measurements, such as body fat percentage and waist-to-hip ratio, can contribute to a more comprehensive understanding of an individual's health. Additionally, advancements in technology, such as 3D body scanning and other imaging techniques, may offer more precise alternatives to traditional BMI calculations.

REFERENCES

- Dataset: <https://www.kaggle.com/datasets/burnoutminer/heights-and-weights-dataset>
- Information Source: https://en.wikipedia.org/wiki/Body_mass_index
- https://www.cdc.gov/healthyweight/assessing/bmi/adult_bmi/index.html
- <https://www.cdc.gov/healthyweight/assessing/bmi/index.html>